

09 Orthogonality



9.2 3×3 Orthogonal Matrices

A 3D geometric view of orthogonal matrices: orthonormal bases, the Gram matrix, orthogonal projection, and orthogonal complements

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

The identity, rotation, reflection, and permutation matrices — and their combinations — are all orthogonal matrices

3

Use matrix multiplication's outer-product view to combine projection matrices into the identity matrix

5

The orthogonal complement of a subspace contains every vector perpendicular to it

7

Project orthogonally onto coordinate planes (x_2x_3 , x_1x_3 , x_1x_2) and understand how a 1D subspace and a 2D subspace form orthogonal complements

2

Use matrix multiplication's inner-product view to expand an orthogonal matrix's Gram matrix

4

Understand "two-step projection": scalar projection first, then vector projection

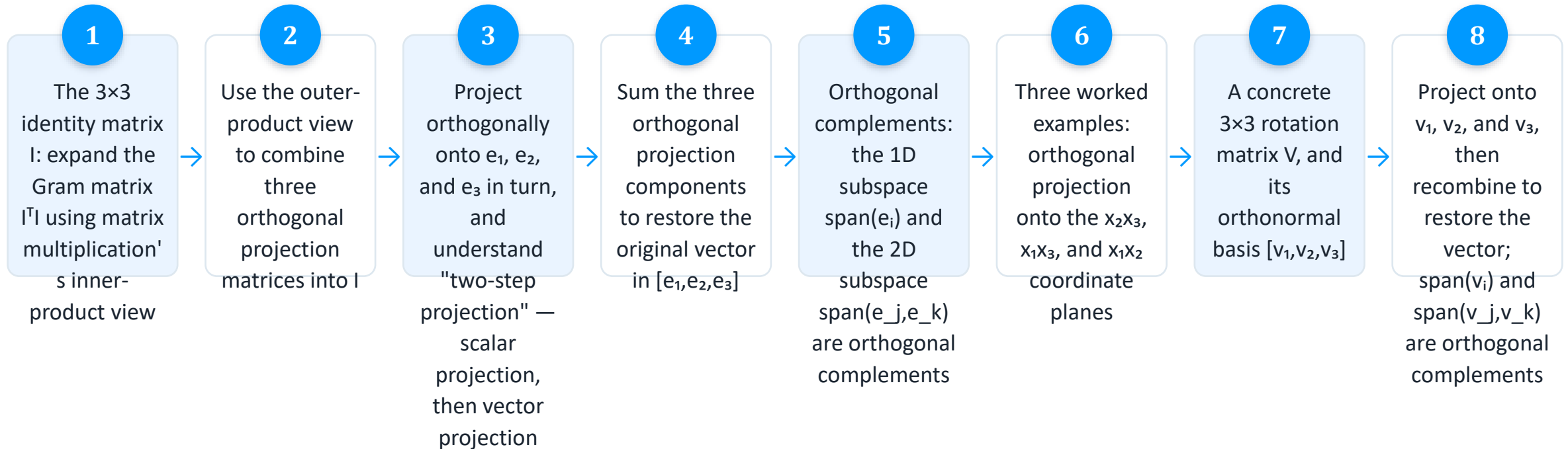
6

In 3D, use the standard orthonormal basis $[e_1, e_2, e_3]$ and an orthonormal basis $[v_1, v_2, v_3]$ to decompose and recombine a vector

8

A 3×3 rotation matrix is also an orthogonal matrix, with columns forming an orthonormal basis

From a 3D Identity Matrix to a 3D Rotation Matrix — Understanding 3×3 Orthogonal Matrices



Seven Lenses for Every Topic in This Chapter

1 Geometric view: the identity, rotation, reflection, permutation matrices, and their combinations are all orthogonal

2 Algebraic property: $\mathbf{Q}^T = \mathbf{Q}^{-1}$ — transpose equals inverse; the determinant is always +1 or -1

3 Gram-matrix view: the columns are pairwise orthogonal, and each column is a unit vector

4 Projection view: using an orthogonal matrix's columns to project — the sum of projections restores the vector

5 Orthogonal-complement view: every subspace has an orthogonal complement perpendicular to it everywhere

6 How Gram–Schmidt orthogonalization and QR decomposition build an orthonormal basis from any basis

7 Where orthogonal matrices show up: regression, spectral decomposition, SVD, principal component analysis

The 3×3 identity matrix I Is Also an Orthogonal Matrix

$$I_{3 \times 3} @ I_{3 \times 3}^T = I_{3 \times 3}^T @ I_{3 \times 3} = I_{3 \times 3}$$

- The last section took a planar view of 2×2 orthogonal matrices; this section "raises the dimension" to 3D space
- Start with the Gram matrix $I^T @ I$, expanded using matrix multiplication's first view — the inner-product view

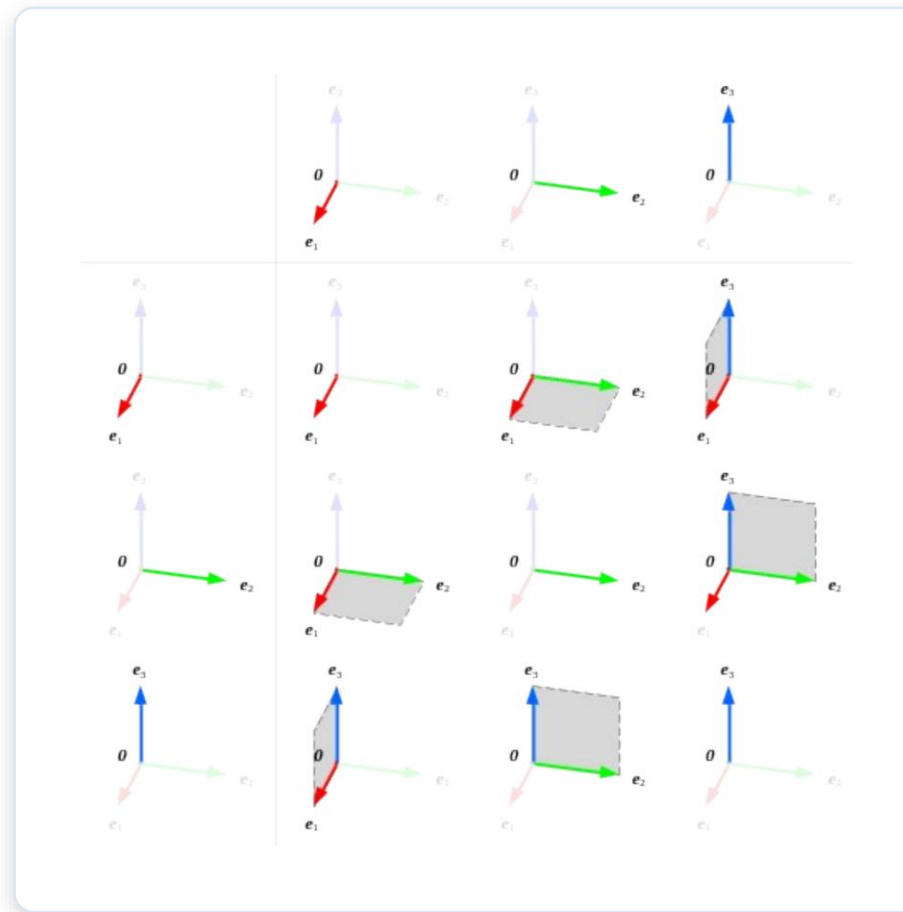
Write I as $[e_1, e_2, e_3]$ and expand using the inner-product view

$$I^T @ I = [e_1 \cdot e_1, e_1 \cdot e_2, e_1 \cdot e_3; e_2 \cdot e_1, e_2 \cdot e_2, e_2 \cdot e_3; e_3 \cdot e_1, e_3 \cdot e_2, e_3 \cdot e_3] = I_{3 \times 3}$$

- The diagonal entries are e_1, e_2, e_3 dotted with themselves — the squared L^2 norm — each equal to 1
- The off-diagonal entries are pairwise inner products; since the standard basis vectors are mutually perpendicular, each equals 0

Figure 1 · Expanding the Identity Matrix's Gram Matrix via the Inner-Product View

- 9 geometric combinations correspond one-to-one with the 9 entries of the Gram matrix: diagonal entries are single basis vectors; off-diagonal entries are parallelograms spanned by two basis vectors



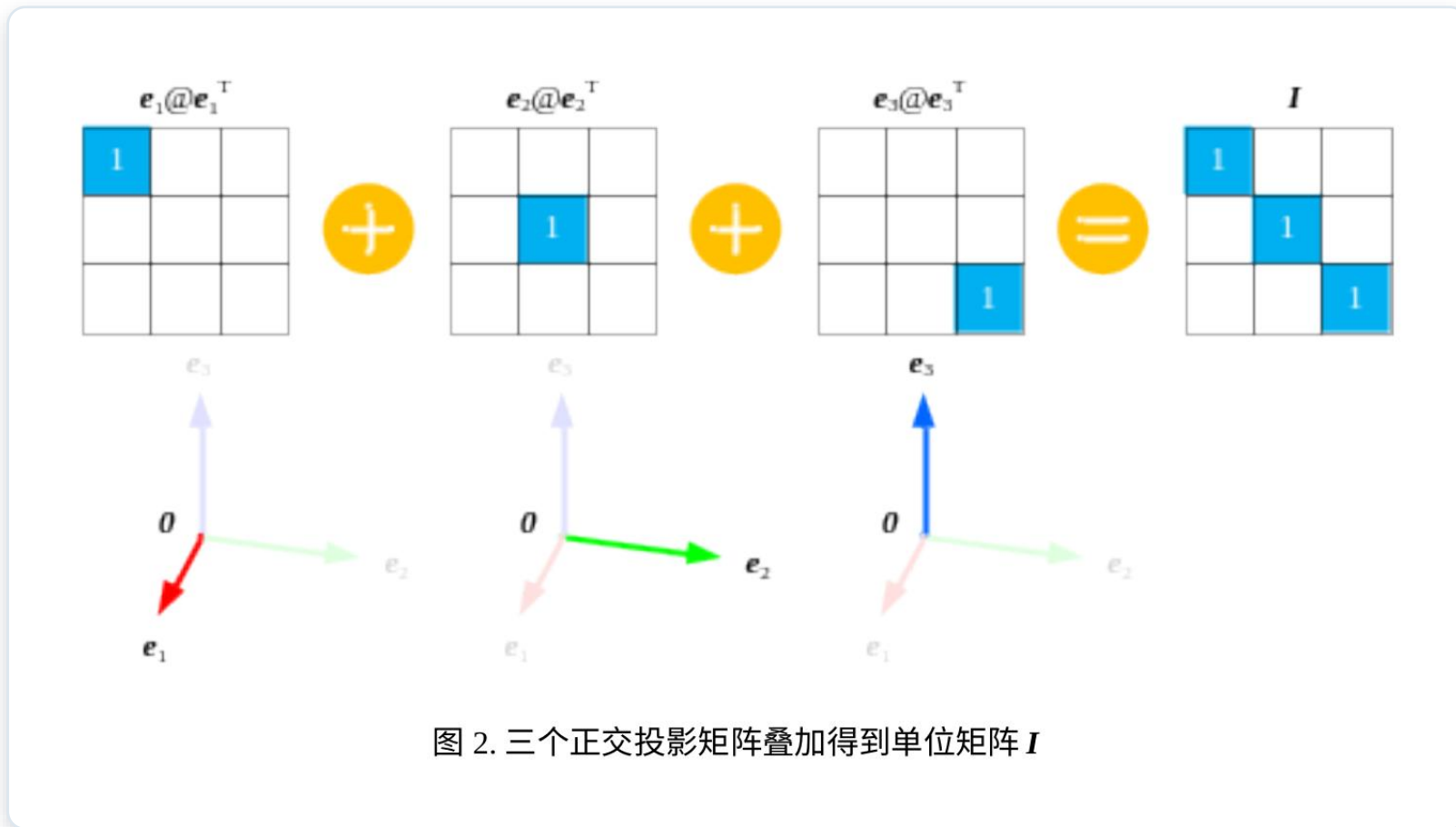
Expand with the outer-product view — a sum of outer products restores the identity matrix

$$[e_1 \ e_2 \ e_3] @ [e_1^T; e_2^T; e_3^T] = e_1 @ e_1^T + e_2 @ e_2^T + e_3 @ e_3^T = I_{3 \times 3}$$

- What we see is a sum of three orthogonal projection matrices, whose result is the identity matrix
- $e_1 @ e_1^T @ x$ projects vector x onto the x_1 axis; $e_2 @ e_2^T @ x$ projects onto the x_2 axis, and so on

Figure 2 · Three Orthogonal Projection Matrices Combine into the Identity Matrix I

- The three rank-one matrices $e_1 @ e_1^T$, $e_2 @ e_2^T$, and $e_3 @ e_3^T$ sum exactly to the identity matrix



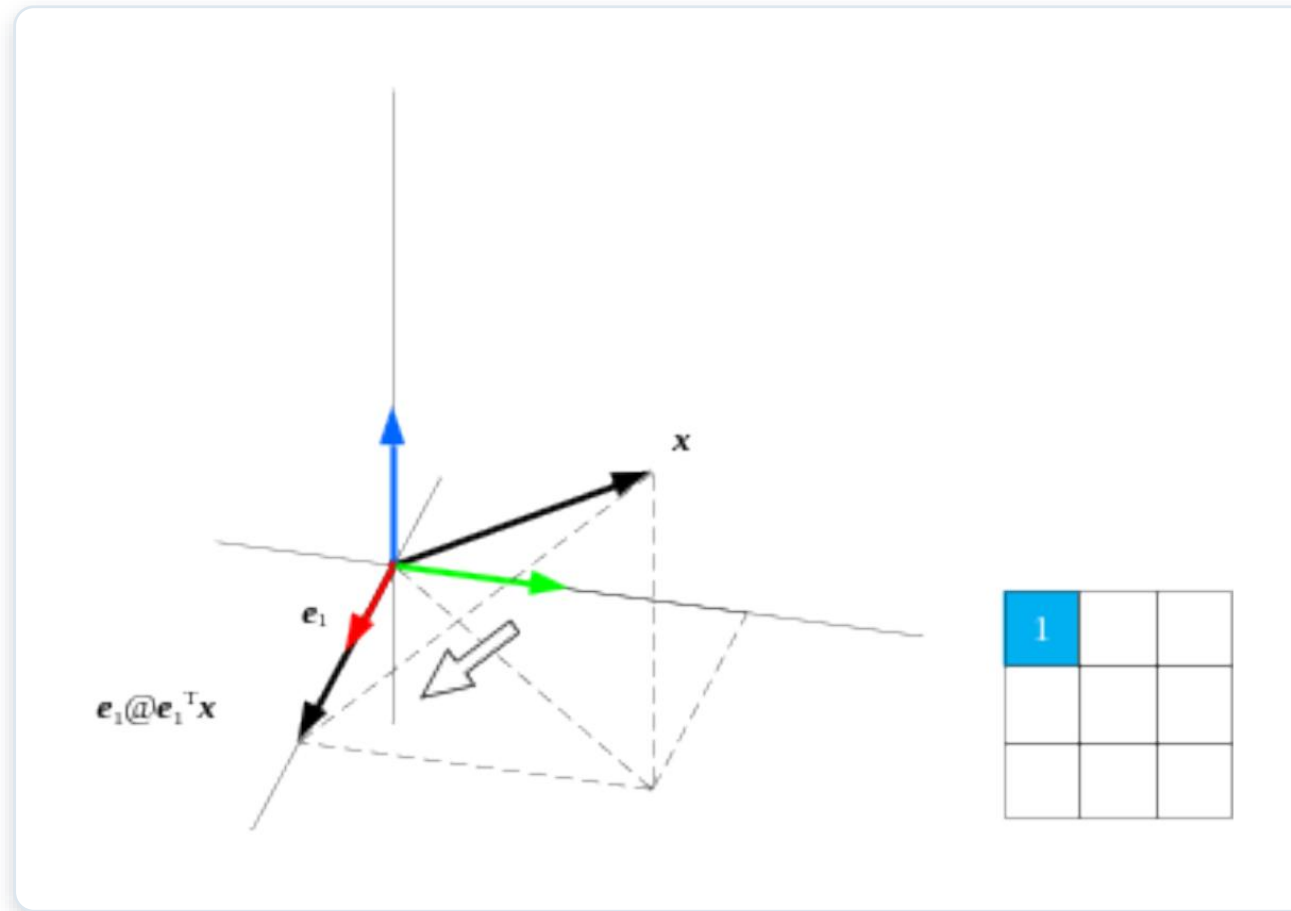
Projecting 3D column vector x onto direction (unit vector) e_1 gives projection matrix P_1

$$P_1 = e_1 @ e_1^T = [1, 0, 0; 0, 0, 0; 0, 0, 0]$$

- Projection matrix P_1 is a rank-one matrix and a symmetric matrix: $(e_1 @ e_1^T)^T = e_1 @ e_1^T$

Figure 3 · Orthogonal Projection of 3D Vector x onto Direction e_1

- The projection result P_1x keeps only the component of x along e_1



"TWO-STEP PROJECTION"

Splitting Orthogonal Projection into Direction and Scalar — This Is "Two-Step Projection"

$$P_1 \mathbf{x} = \mathbf{e}_1 @ (\mathbf{e}_1^T @ \mathbf{x}) \quad \mathbf{x}^T @ \mathbf{e}_1 @ \mathbf{e}_1^T$$

- First project $\mathbf{x}$ onto $\text{span}(\mathbf{e}_1)$ — a scalar projection $(\mathbf{e}_1^T @ \mathbf{x})$; then describe the result back in 3D — a vector projection in the direction $\mathbf{e}_1$
- Transposing both sides gives a row vector $\mathbf{x}^T$, representing a point in 3D space

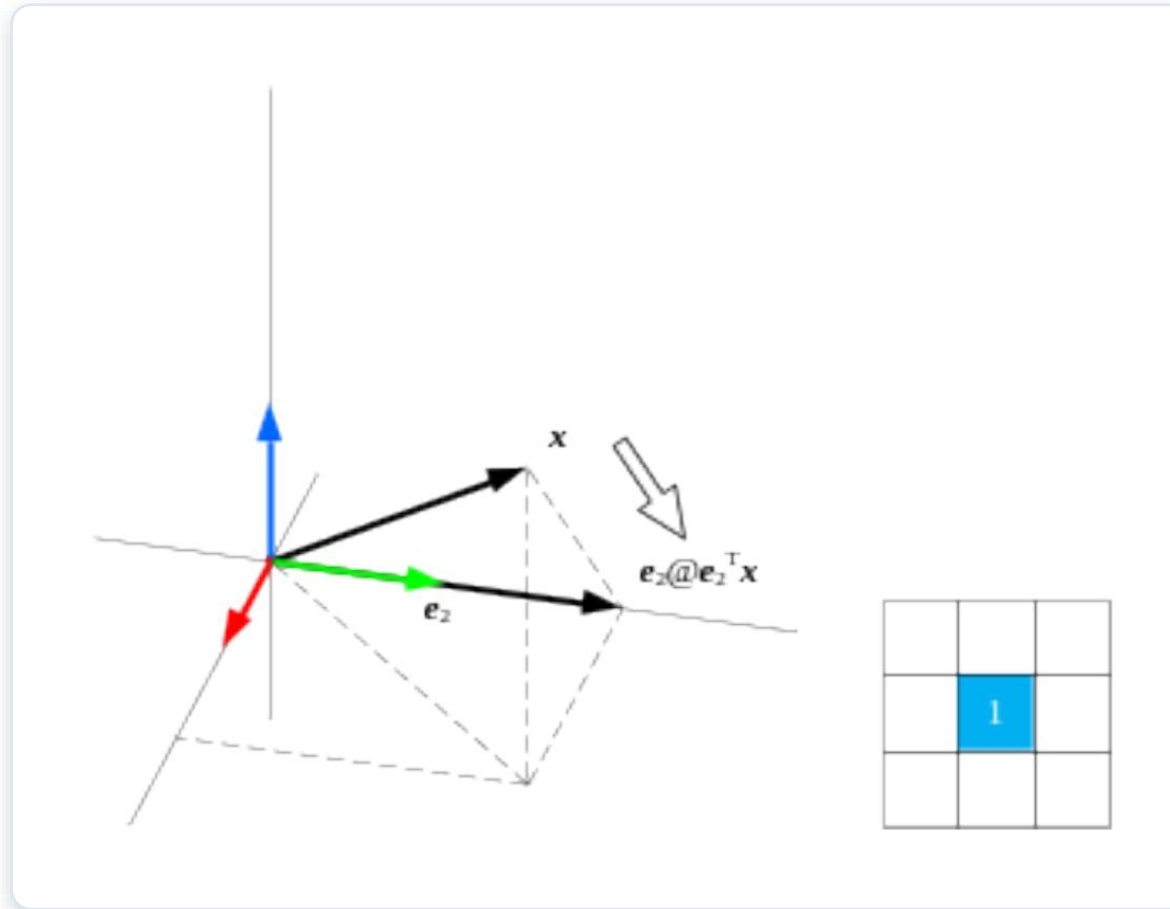
Projecting 3D column vector $\mathbf{x}$ onto $\mathbf{e}_2$ gives projection matrix P_2

$$P_2 = \mathbf{e}_2 \mathbf{e}_2^T = [0,0,0; 0,1,0; 0,0,0]$$

- Projection matrix P_2 is likewise a rank-one, symmetric matrix
- The coordinate of column vector $\mathbf{x}$ in $\text{span}(\mathbf{e}_2)$ is exactly the scalar projection $\mathbf{e}_2^T \mathbf{x} = x_2$

Figure 4 · Orthogonal Projection of 3D Vector x onto Direction e_2

- The projection result P_2x keeps only the component of x along e_2



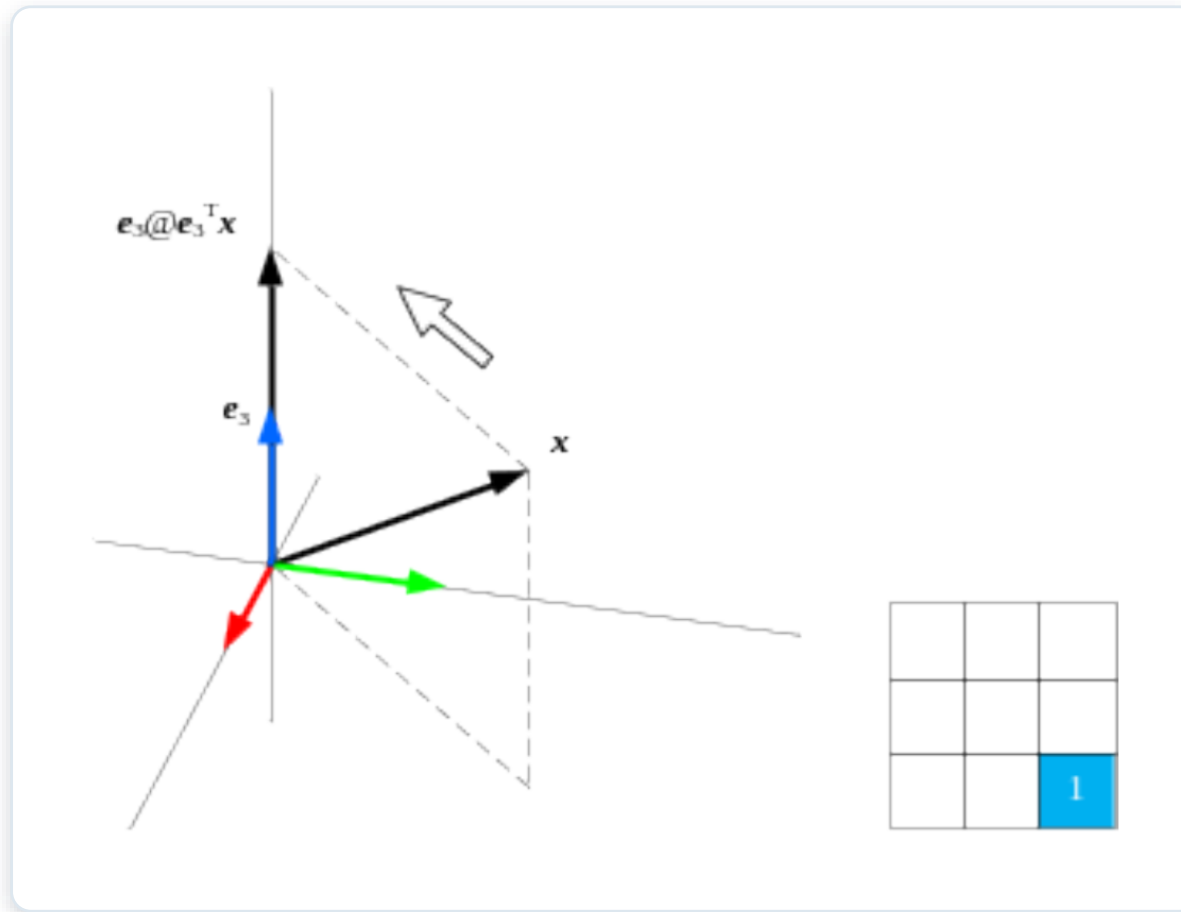
Projecting 3D column vector $\mathbf{x}$ onto direction $\mathbf{e}_3$ gives projection matrix P_3

$$P_3 = \mathbf{e}_3 \mathbf{e}_3^T = [0,0,0; 0,0,0; 0,0,1]$$

- The coordinate of column vector $\mathbf{x}$ in $\text{span}(\mathbf{e}_3)$ is exactly the scalar projection $\mathbf{e}_3^T \mathbf{x} = x_3$

Figure 5 · Orthogonal Projection of 3D Vector x onto Direction e_3

- The projection result P_3x keeps only the component of x along e_3



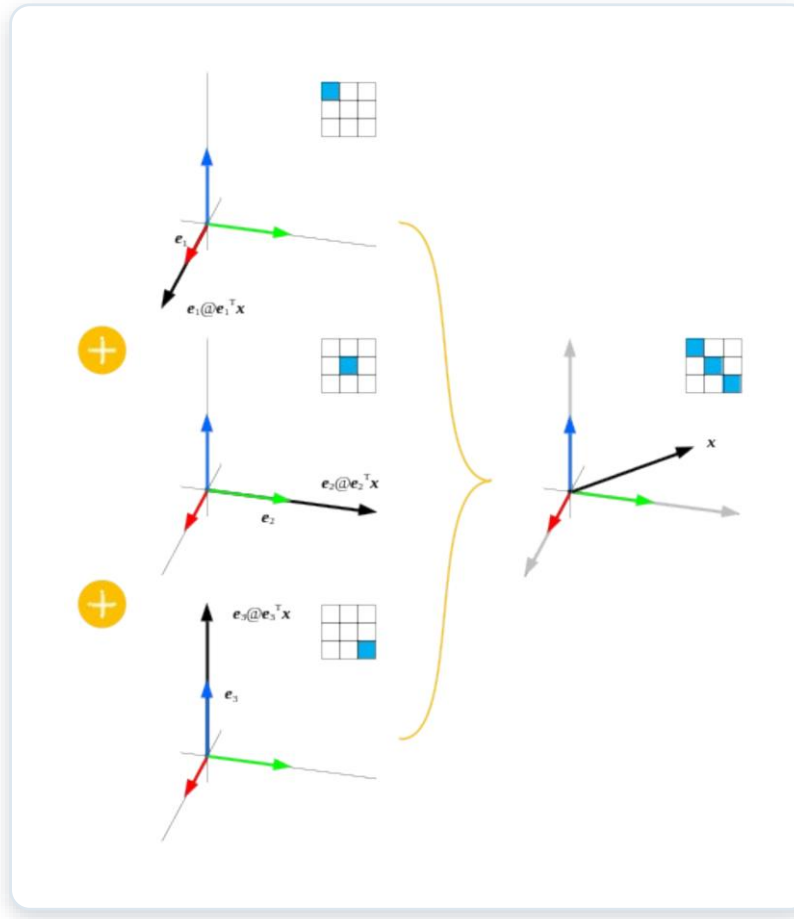
P_1x , P_2x , and P_3x Each Hold Only Part of x 's Information; Summing Them

$$P_1x + P_2x + P_3x = (P_1 + P_2 + P_3)x = Ix$$

- We restore all of x 's information in $[e_1, e_2, e_3]$

Figure 6 · Restoring 3D Column Vector $\mathbf{x}$ in $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$

- Summing the three orthogonal projection components restores the original vector $\mathbf{x}$ exactly, with no loss



Rearranging $P_1 + P_2 + P_3 = I$ Gives Three Interesting Identities

$$I - e_1 @ e_1^T = e_2 @ e_2^T + e_3 @ e_3^T \quad I - e_2 @ e_2^T = e_1 @ e_1^T + e_3 @ e_3^T \quad I - e_3 @ e_3^T = e_1 @ e_1^T + e_2 @ e_2^T$$

- What do these three matrices represent? Do you recognize the orthogonal projection matrix defined by a normal vector?

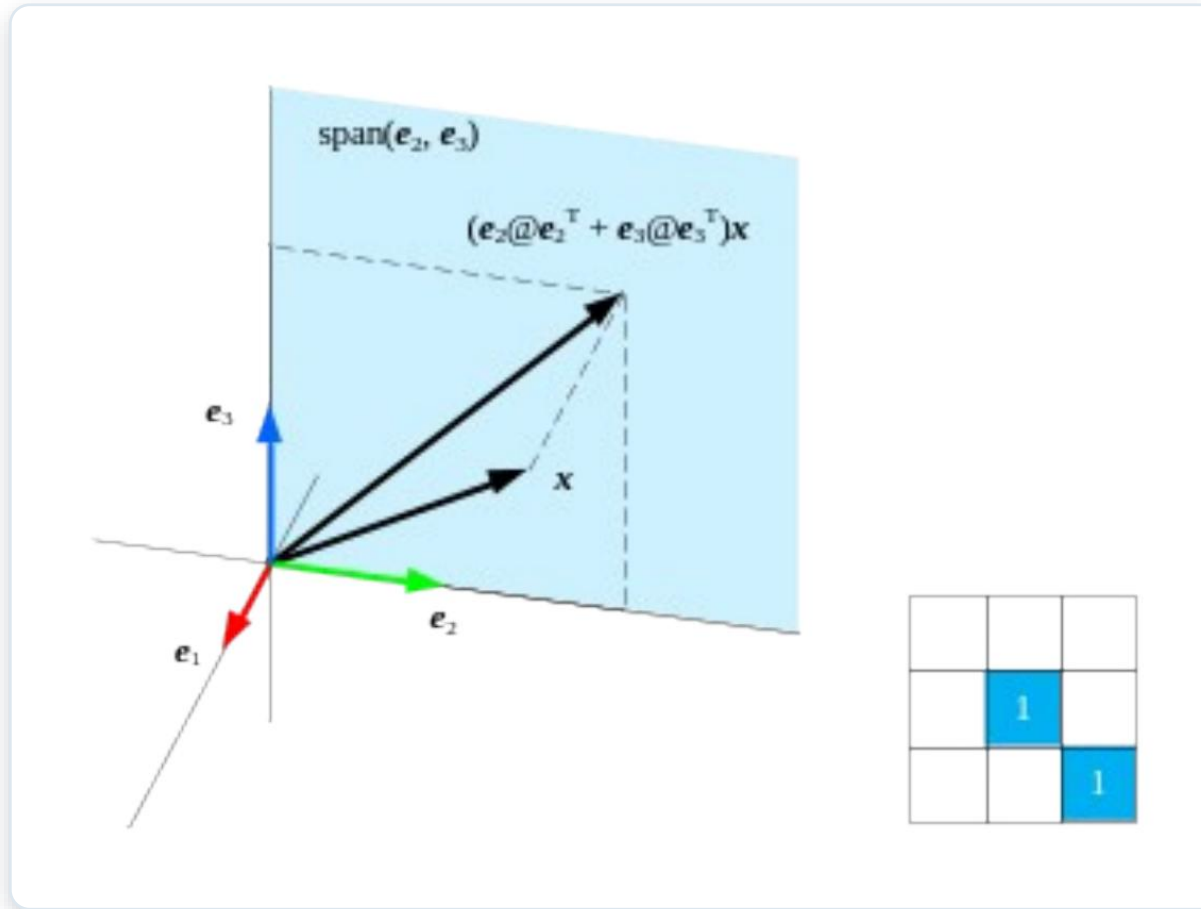
The normal vector of the x_2x_3 plane is e_1 ; the orthogonal complement equals the sum of two axis projections

$$(I - e_1 @ e_1^T) x = (e_2 @ e_2^T + e_3 @ e_3^T) x = e_2 @ e_2^T @ x + e_3 @ e_3^T @ x$$

- The right side means projecting onto the x_2 axis and the x_3 axis separately, then summing
- $\text{span}(e_1)$ and $\text{span}(e_2, e_3)$ are orthogonal complements

Figure 7 · Orthogonal Projection of a 3D Column Vector onto the x_2x_3 Plane

- Projecting onto the 1D space $\text{span}(e_1)$ and the 2D space $\text{span}(e_2, e_3)$ each yields only partial information; together they recover the whole



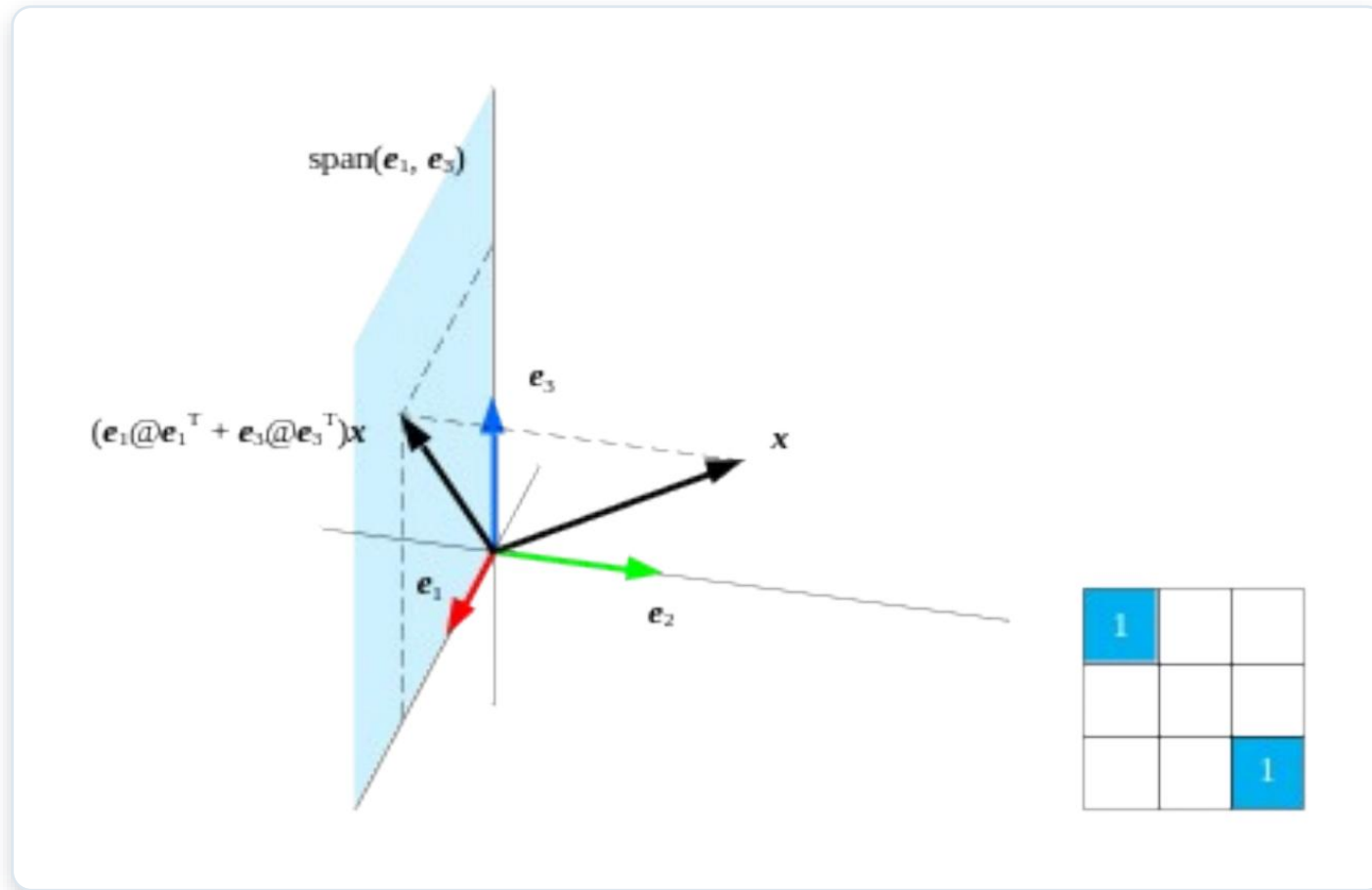
The normal vector of the x_1x_3 plane is e_2 ; the orthogonal complement equals the sum of two axis projections

$$(I - e_2 @ e_2^T) x = e_1 @ e_1^T @ x + e_3 @ e_3^T @ x$$

- The right side means projecting onto the x_1 axis and the x_3 axis separately, then summing
- $\text{span}(e_2)$ and $\text{span}(e_1, e_3)$ are orthogonal complements

Figure 8 · Orthogonal Projection of a 3D Column Vector onto the x_1x_3 Plane

- Projecting onto two mutually orthogonal-complement spaces each yields only partial information; together they recover the whole



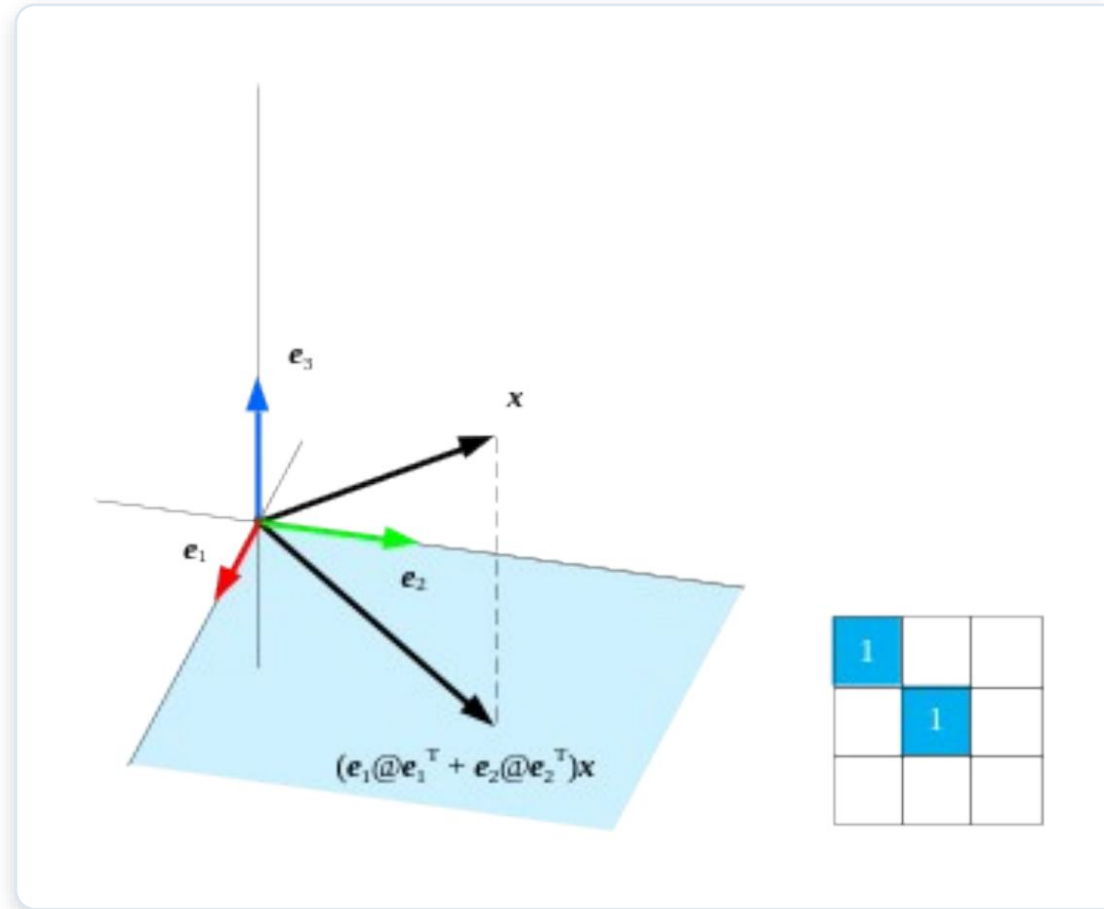
Try it yourself: the x_1x_2 plane's normal vector is e_3

$$(I - e_3 e_3^T)x = e_1 e_1^T x + e_2 e_2^T x$$

- $\text{span}(e_3)$ and $\text{span}(e_1, e_2)$ are orthogonal complements

Figure 9 · Orthogonal Projection of a 3D Column Vector onto the x_1x_2 Plane

- These three coordinate-plane examples show the general pattern: a 1D subspace and a 2D subspace are always orthogonal complements



Given the following 3D rotation matrix (about the origin) V

$$V = [4/5, -3/5, 0; 3/5, 4/5, 0; 0, 0, 1] = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$$

- This is a rotation about the x_3 axis: $\mathbf{v}_1 = [4/5, 3/5, 0]^T$, $\mathbf{v}_2 = [-3/5, 4/5, 0]^T$, $\mathbf{v}_3 = [0, 0, 1]^T$
- Matrix V is also orthogonal, satisfying $V @ V^T = V^T @ V = I$

First, the Gram matrix — write V as $[v_1, v_2, v_3]$ and expand using the inner-product view

$$V^T @ V = [v_1 \cdot v_1, v_1 \cdot v_2, v_1 \cdot v_3; v_2 \cdot v_1, v_2 \cdot v_2, v_2 \cdot v_3; v_3 \cdot v_1, v_3 \cdot v_2, v_3 \cdot v_3] = I_{3 \times 3}$$

- The diagonal entries are V 's columns dotted with themselves — the squared L^2 norm — each equal to 1
- The off-diagonal entries are pairwise inner products of V 's columns, each equal to 0, because V 's columns are pairwise orthogonal

Expand with the outer-product view — a sum of outer products restores the identity matrix

$$V@V^T = \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T = \mathbf{I}_{3 \times 3}$$

- Once again, this is a sum of three orthogonal projection matrices, whose result is the identity matrix

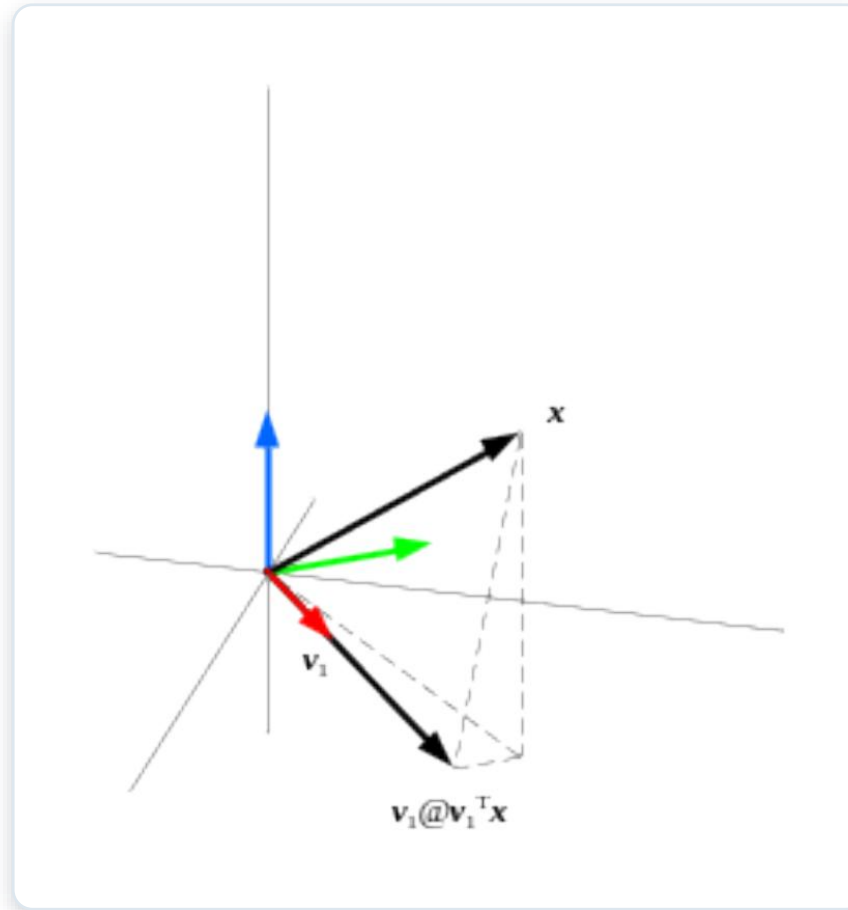
Projecting 3D column vector x onto direction (unit vector) v_1 gives projection matrix P_1

$$P_1 = v_1 v_1^T = \begin{bmatrix} 16/25 & 12/25 & 0 \\ 12/25 & 9/25 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- This projection matrix is a rank-one, symmetric matrix
- The coordinate of column vector x in $\text{span}(v_1)$ — the scalar projection — is $v_1^T x = 4x_1/5 + 3x_2/5$

Figure 10 · Orthogonal Projection of 3D Vector x onto Direction v_1

- Even though the orthonormal basis has been rotated, orthogonal projection is built exactly the same way as with the standard basis



Likewise, projecting 3D column vector x onto v_2 and v_3 gives projection matrices P_2 and P_3

$$P_2 = v_2 @ v_2^T = [9/25, -12/25, 0; -12/25, 16/25, 0; 0, 0, 0] \quad P_3 = v_3 @ v_3^T = [0, 0, 0; 0, 0, 0; 0, 0, 1]$$

- The coordinates of column vector x in $\text{span}(v_2)$ and $\text{span}(v_3)$ are the scalar projections $v_2^T @ x$ and $v_3^T @ x$

Summing P_1x , P_2x , and P_3x Restores All of x 's Information

$$(\mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x} = \mathbf{I} @ \mathbf{x} = \mathbf{x}$$

- Whether using the standard basis or a rotated orthonormal basis, the sum of orthogonal projections always restores the original vector losslessly

Rearranging Again Gives Three More Interesting Identities

$$I - \mathbf{v}_1 @ \mathbf{v}_1^T = \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T \quad I - \mathbf{v}_2 @ \mathbf{v}_2^T = \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_3 @ \mathbf{v}_3^T \quad I - \mathbf{v}_3 @ \mathbf{v}_3^T = \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T$$

- First identity: the left side means projecting onto the $\text{span}(\mathbf{v}_2, \mathbf{v}_3)$ plane, whose normal vector is $\mathbf{v}_1$

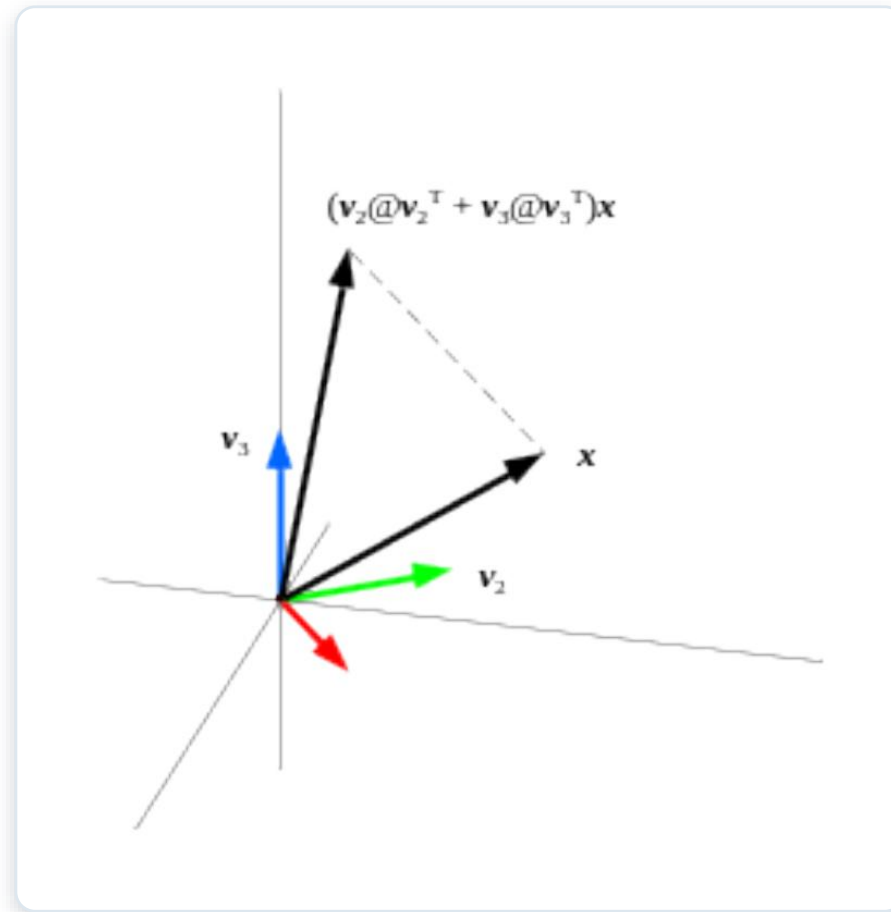
The Right Side Means Projecting onto $\text{span}(\mathbf{v}_2)$ and $\text{span}(\mathbf{v}_3)$ Separately, Then Summing

$$(I - \mathbf{v}_1 @ \mathbf{v}_1^T) \mathbf{x} = \mathbf{v}_2 @ \mathbf{v}_2^T @ \mathbf{x} + \mathbf{v}_3 @ \mathbf{v}_3^T @ \mathbf{x}$$

- $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_2, \mathbf{v}_3)$ are orthogonal complements

Figure 11 · Orthogonal Projection of 3D Vector x onto $\text{span}(v_2, v_3)$

- This plane passes through the origin with normal vector v_1



Second identity: the left side means projecting onto $\text{span}(\mathbf{v}_1, \mathbf{v}_3)$, whose normal vector is $\mathbf{v}_2$

$$(I - \mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{x} = \mathbf{v}_1 @ \mathbf{v}_1^T @ \mathbf{x} + \mathbf{v}_3 @ \mathbf{v}_3^T @ \mathbf{x}$$

- The right side means projecting onto $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_3)$ separately, then summing
- $\text{span}(\mathbf{v}_2)$ and $\text{span}(\mathbf{v}_1, \mathbf{v}_3)$ are orthogonal complements

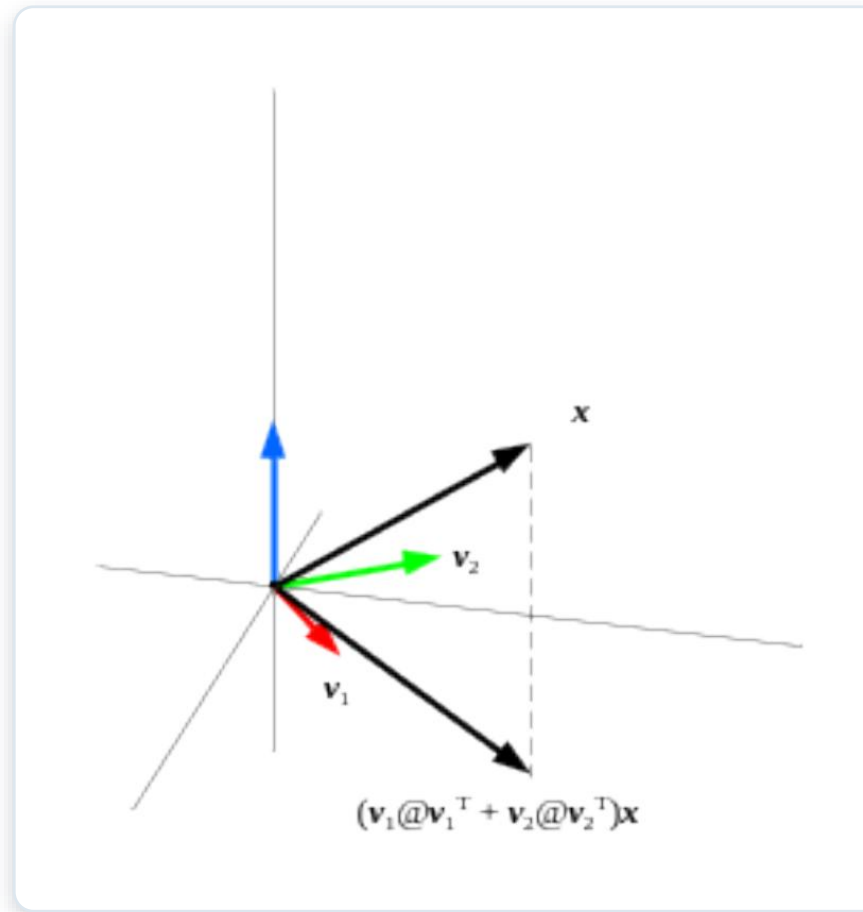
Try the third identity yourself: the left side projects onto $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$, with normal vector $\mathbf{v}_3$

$$(I - \mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x} = \mathbf{v}_1 @ \mathbf{v}_1^T @ \mathbf{x} + \mathbf{v}_2 @ \mathbf{v}_2^T @ \mathbf{x}$$

- $\text{span}(\mathbf{v}_3)$ and $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ are orthogonal complements

Figure 12 · Orthogonal Projection of 3D Vector $\mathbf{x}$ onto $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$

- This plane passes through the origin with normal vector $\mathbf{v}_3$



What Does a 3D View of Orthogonal Matrices Reveal?

Both the 3×3 identity matrix and the rotation matrix are orthogonal: their columns form an orthonormal basis

The Gram matrix $V^T V = I$ is the core tool for judging an orthonormal basis or an orthogonal matrix

"Two-step projection": scalar projection (inner product) first, then vector projection (outer product) — together they form the orthogonal projection matrix

General pattern for orthogonal complements: a 1D subspace $\text{span}(v)$ and the 2D subspace spanned by the other two basis vectors are orthogonal complements

Glossary

Orthogonal Matrix 正交矩阵 — a square matrix satisfying $Q^T @ Q = Q @ Q^T = I$

Orthonormal Basis 规范正交基 — a set of vectors that are pairwise orthogonal and unit length

Orthogonal Projection 正交投影 — $P = u @ u^T$, the projection of a vector onto unit vector u

Orthogonal Complement 正交补 — the space of all vectors perpendicular to every vector in a given subspace

Rotation Matrix 旋转矩阵 — an orthogonal matrix with determinant +1 that preserves length and angle

Standard Orthonormal Basis 标准正交基 — $[e_1, e_2, e_3]$, the unit vectors along the coordinate axes

Gram Matrix 格拉姆矩阵 — $V^T @ V$, diagonal holds squared lengths, off-diagonal holds pairwise inner products

Two-Step Projection 二次投影 — scalar projection first ($u^T @ x$), then vector projection (u times that scalar)

Rank-One Matrix 秩一矩阵 — a matrix formed by the outer product of a single vector, such as $u @ u^T$

span 张成空间 — the space spanned by the linear combinations of a set of vectors

EXPLORE FURTHER

Further Reading

Recap · 9.1 2×2 Orthogonal Matrices

Previous section — a planar geometric view of orthogonal matrices

Further Reading · 9.3 Properties of Orthogonal Matrices

Later in this chapter — a systematic look at algebraic properties

Further Reading · 9.4 Gram–Schmidt Orthogonalization

Later in this chapter — building orthogonal bases using projection ideas

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Matrices A – I (used in Q1 – Q2)

Label	Matrix	Label	Matrix
A	[1,0,0; 0,1,0; 0,0,1]	F	[1,0; 0,0; 0,1] (3×2)
B	[1,0; 0,1; 0,0] (3×2)	G	[1,0,0; 0,-1,0; 0,0,-1]
C	[1,0,1; 0,1,0; 0,0,1]	H	[1,0,0; 0,0,1; 0,1,0]
D	[1,0,0; 0,1,0; 0,0,-1]	I	[√2/2,-√2/2,0; √2/2,√2/2,0; 0,0,1]
E	[1,0; 0,-1; 0,0] (3×2)		

B, E, and F are all 3×2 non-square matrices — even with orthonormal columns, a non-square matrix cannot be called orthogonal; judge each case using this section's ideas.

Q1 – Q9

Q1 Determine which of the matrices $A - I$ in the table above are orthogonal.

Q3 Compute the determinant of each orthogonal matrix in Q1, and observe which are rotations and which are reflections.

Q5 Compute the scalar and vector projections of the 3D column vector $[1, 2, 3]^T$ onto e_1 , e_2 , and e_3 .

Q7 Compute the orthogonal projection of the 3D column vector $[1, 2, 3]^T$ onto the x_2x_3 , x_1x_3 , and x_1x_2 planes.

Q9 Project $[1, 2, 3]^T$ onto v_1 , v_2 , and v_3 separately, then combine the results and verify they restore the original vector.

Q2 Geometrically, what transform does each orthogonal matrix in Q1 correspond to?

Q4 Verify that although matrix B (3×2) has two orthonormal columns, $B @ B^T$ does not equal the identity — a non-square matrix is not orthogonal.

Q6 Project $[1, 2, 3]^T$ onto e_1 , e_2 , and e_3 separately, then combine the results and verify they restore the original vector.

Q8 Given rotation matrix $V = [4/5, -3/5, 0; 3/5, 4/5, 0; 0, 0, 1]$, compute the projection matrices for v_1 , v_2 , and v_3 respectively.

Q10 – Q17

Q10 Write a Python function `is_orthogonal(M)` that checks whether a square matrix is orthogonal (verify $M @ M.T$ is approximately I).

Q12 Write a Python function `gram_matrix(V)` that computes the Gram matrix of a matrix's column vectors, and verify an orthogonal matrix's Gram matrix is I .

Q14 Use matplotlib 3D plotting to visualize summing the orthogonal projections of a 3D vector onto e_1 , e_2 , and e_3 .

Q16 Verify with code: I minus the projection matrix onto one coordinate axis equals the sum of the projection matrices onto the other two axes' plane.

Q11 Use numpy to check which of the 9 matrices in the table above satisfy the orthogonal-matrix definition, and compare with your hand calculations.

Q13 Write a Python function `axis_projection_matrix(i, n)` that returns the projection matrix onto the i -th coordinate axis in n -dimensional space.

Q15 Write a Python function `plane_projection_matrix(normal)` that returns the projection matrix onto the plane with the given normal vector.

Q17 Explore: given an arbitrary unit vector v_1 , how do you construct v_2 and v_3 so that $[v_1, v_2, v_3]$ forms a 3×3 orthogonal matrix?

Q18 – Q25

Q18 Write a Python function `orthogonal_projection_3d(x, u)` that computes the orthogonal projection of 3D vector x onto unit vector u .

Q20 Explore: given orthonormal basis $[v_1, v_2, v_3]$, does the sum of x 's projections onto the three directions also always equal x ? Verify with code.

Q22 Explore $\text{span}(e_1)$ and $\text{span}(e_2, e_3)$ as orthogonal complements: verify with code that any vector from each subspace has zero inner product with any vector from the other.

Q24 Explore: change this section's rotation matrix V (about the x_3 axis) to an arbitrary angle θ , and observe how v_1 and v_2 trace out paths as θ varies.

Q19 Verify with code: for any 3D vector x , the sum of its projections onto e_1 , e_2 , and e_3 always equals x .

Q21 Write code comparing vector x 's coordinates under the standard basis $[e_1, e_2, e_3]$ versus the rotated basis $[v_1, v_2, v_3]$.

Q23 Comprehensive: for any 3×3 orthogonal matrix Q , use Python to verify its three columns are pairwise orthogonal and each has unit length.

Q25 Preview thought: the next section systematically covers the properties of orthogonal matrices — does the "transpose equals inverse" property seen here hold for every orthogonal matrix?

10 Prompts to Explore With AI

1 Explain why expanding the 3×3 identity matrix's $I^T I$ gives 1s on the diagonal and 0s off the diagonal.

2 Explain what "two-step projection" means: why can orthogonal projection be split into scalar projection and vector projection?

3 Explain the geometric meaning behind $e_1 @ e_1^T + e_2 @ e_2^T + e_3 @ e_3^T = I$.

4 Explain why $\text{span}(e_1)$ and $\text{span}(e_2, e_3)$ are orthogonal complements, and whether this pattern extends to n dimensions.

5 Use Python to demonstrate: projecting a 3D vector onto three orthogonal directions, then combining the results to restore the original vector.

10 Prompts to Explore With AI

6 Explain why the columns of the 3×3 rotation matrix V are pairwise orthogonal and each has unit length.

7 Explain which views of matrix multiplication correspond to the Gram matrix identity $V^T V = I$ and the outer-product-sum identity $V V^T = I$.

8 Explain why the projection matrix onto a coordinate plane (such as the $x_2 x_3$ plane) can be written as I minus the projection matrix onto its normal vector.

9 Explain why an orthogonal matrix's columns must be both unit length and pairwise orthogonal — is either condition alone enough?

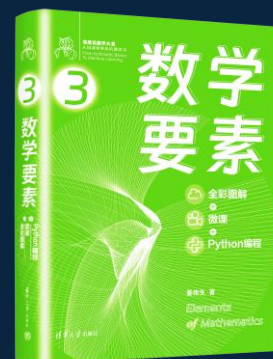
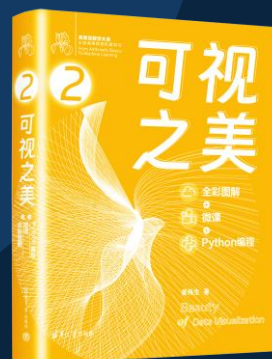
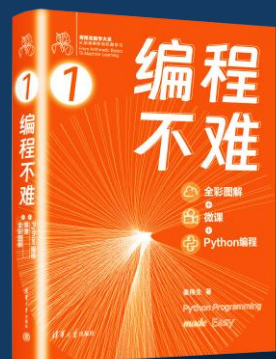
10 Explain what applications 3×3 orthogonal matrices might have in 3D graphics and robot pose representation.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

